Explanations That Arrive in Time: Deadline-Aware, Control-Loop-Structured Explainable AI for Intrusion Detection in Cyber-Physical Systems

[Author names withheld for review]

[Affiliation]

Abstract

Explainable AI (XAI) is widely proposed to help operators act on alarms of machine-learning intrusion detectors in cyber-physical systems (CPS). In a CPS, an explanation is only useful if it is correct and arrives while the physical process can still be protected. Prior work has shown that attributions computed when the alarm fires are often wrong, and reported explanation latencies are never related to how long the physical process can tolerate an attack. We argue that XAI for CPS should be built on, and evaluated against, the physical structure of the system: its control loops and its safety deadlines. We frame explanation as *identifying the compromised control loop before a physically derived deadline*, and introduce deadline-conditioned explanation metrics that combine detection delay, explainer latency and the correctness of the delivered explanation. We propose LX, an exact Shapley attribution over control loops with an incident-local reference and a fixed, data-independent cost of 2^L detector queries, together with temporal evidence accumulation (TEA), a method-agnostic aggregation of attributions over an incident. We evaluate on industrial process CPS, the CPS class with public attack ground truth: we curate loop-level ground truth for all 50 attacks of the public HAI 21.03 hardware-in-the-loop testbed and release PCL-Sim, a closed-loop simulator whose 280 attacks carry physical hazard deadlines. We compare 12 attribution configurations across four detector families with leakage-free protocols and paired statistical tests. Single-shot attributions of every method degrade sharply as attacks propagate after the alarm; on HAI, LX falls from 0.79 top-1 accuracy at the alarm to 0.30 after 180 s, and TEA restores it to 0.64. Grouping by control loops is essential: correlation-based or random groupings of equal size fail. LX is the most accurate black-box method at the

alarm on HAI (0.79), but with TEA several methods, including integrated gradients, are statistically tied. Under sensor noise and a non-adaptive decoy, LX+TEA offers no robustness advantage over the baselines. On hazardous simulated attacks, 86 % of deadline misses are due to late or missed detection rather than wrong explanations. A random window split inflates supervised detection F1 on HAI from 0.575 to 0.940.

Keywords: cyber-physical systems, explainable AI, intrusion detection, Shapley values, control loops, real-time explanation, process safety time

1. Introduction

Cyber-physical systems (CPS) couple computation and communication with physical processes through sensing, control and actuation. Attacks on them have physical consequences, and machine-learning anomaly detectors are a widely studied line of defence. A forecaster learns the normal dynamics of sensors, setpoints and actuator commands, and raises an alarm when observations depart from its predictions (Kravchik and Shabtai, 2018; Deng and Hooi, 2021; Fung et al., 2024). An alarm alone, however, does not tell an operator *what* to do. To respond, the operator must know which part of the physical system has been manipulated. The natural unit of a CPS is the feedback control loop: a reference or setpoint (SP), a measured process variable (PV) and a control output (CV) tied together by one controller. It is also the unit in which attacks on industrial CPS are specified (Shin et al., 2021). Explainable AI (XAI) is the usual answer: post-hoc attribution methods such as SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016) or integrated gradients (Sundararajan et al., 2017) rank input features by their contribution to the alarm.

Two observations motivate this work. First, the most systematic study to date of attribution for CPS anomaly detection, on industrial control systems (ICS) (Fung et al., 2024) found that attributions computed at the moment an alarm fires—the only moment that matters operationally—are often wrong. The authors went as far as suggesting that attribution may be better suited to non-real-time workflows. Second, the growing literature that does measure the cost of explanations in cyber-physical settings reports it in absolute terms: seconds per explanation on a GPU (Oinonen and Morsi, 2026), time per window against a one-second budget (Franco de la Peña et al., 2026), or milliseconds on an edge device (Ericson et al., 2025). None of

these relates explanation time, or the accuracy of the explanation at the time it is delivered, to what actually bounds the operator's response: the time the physical process can tolerate the attack before reaching an unsafe state. Process-safety engineering formalises this interval as the *process safety time* (International Electrotechnical Commission, 2016).

We argue that the right question is not "how accurate is the explanation?" nor "how fast is it?" but: *does the operator receive a correct localization before the deadline imposed by the process?* This is the question that distinguishes XAI for CPS from XAI for conventional IT intrusion detection, where no physical clock is running. Answering it requires three ingredients the literature lacks: (i) an evaluation that composes detection delay, explanation latency and the time-varying correctness of the explanation into a single deadline-conditioned quantity; (ii) public ground truth at the level operators act on, together with physically meaningful deadlines; and (iii) an attribution method whose accuracy does not collapse during an incident and whose cost is predictable.

Approach. We attribute the detector's anomaly score to the plant's *control loops* with exact Shapley values (LX). Coalitions are evaluated against an incident-local, last-known-good reference rather than a global background. Grounding the explanation in the control structure of the CPS, rather than in individual input features, makes it both actionable and tractable: with the ten or fewer loops of typical process units, exact computation needs a fixed 2^L batched detector evaluations, which yields deterministic, data-independent latency. We combine this with *temporal evidence accumulation* (TEA): a severity-weighted aggregation of normalised attributions over the successive alarm windows of an incident, with a confidence margin that supports an anytime stopping rule. Within the top-ranked loop, an exact tag-level refinement recovers the manipulated points.

Scope. The method requires only a loop map and black-box access to a detector score, both of which exist for most CPS. Our empirical evidence, however, comes from industrial process CPS, the domain with public attack datasets carrying target ground truth: a real hardware-in-the-loop testbed (HAI) and a process simulator. Other CPS domains (power grids, vehicles, medical devices) are discussed but not evaluated.

Contributions.

1. **A deadline-conditioned evaluation framework** for explanations in CPS intrusion detection. We define deadline-conditioned localization DCL@$k(D)$ and an in-time rate against physically derived hazard deadlines. Both account for detection delay, the measured latency of the explainer and the correctness of the explanation actually delivered (Section 3).
2. **LX and TEA**: loop-structured exact Shapley attribution with an incident-local reference, and a generic temporal accumulation scheme that applies to any attribution method (Section 4).
3. **Two benchmarks with localization ground truth.** The first is a curated loop- and tag-level ground-truth table for all 50 attacks of the public HAI 21.03 dataset (Shin et al., 2021), with documented corrections, cross-checked against an independent transcription (49/50 agreement). The second is PCL-Sim, an open closed-loop process simulator with five interacting PI loops. Its 280 attacks follow the HAI taxonomy, and each carries a hazard deadline computed from the true physical state (Section 5).
4. **A leakage-free, statistically tested comparison** of 12 attribution methods $\times$ 4 detector families $\times$ 3 seeds on both benchmarks. It covers accuracy over time, deadline-conditioned localization, latency and evaluation cost, faithfulness, temporal and perturbation stability, sensor-noise and decoy robustness, and ablations (Section 6). A side experiment quantifies how much a random window split inflates detection results on HAI. Such splits appear in several of the XAI-IDS studies we reviewed.

Summary of findings. The results are mixed, and we report them as such.

- *Timing dominates method choice.* Single-shot attributions of every method degrade after the alarm as the attack propagates through coupled loops. On HAI, LX falls from 0.79 top-1 accuracy at the alarm to 0.30 after 180 s. TEA recovers most of this loss for every method (0.64 for LX) and makes the top-1 output temporally stable.

- *Control-loop structure is essential.* Exact Shapley values over correlation clusters or random partitions of the same size are far less accurate than over control loops.

- LX *is competitive, not dominant.* It is the most accurate black-box method at the alarm on HAI. With TEA, however, it is statistically

tied with integrated gradients and other strong baselines, and on PCL-Sim integrated gradients and a training-background variant are slightly better. Its distinguishing properties are a fixed, data-independent cost and gradient-free determinism.

- *Most deadline misses are detection misses.* On hazardous simulated attacks, 86 % of the deadline failures of LX+TEA occur because the attack is detected too late or not at all.
- *Leakage matters.* On HAI, a random window split inflates supervised detection F1 from 0.575 to 0.940.

2. Related work

Interpretable CPS anomaly detection. Several detectors expose interpretable internals:

- graph-attention weights that point to a deviating sensor (GDN; Deng and Hooi, 2021);
- GNN–GRU hybrids with attention (Tang et al., 2023);
- multi-head attention on ICS traffic (Eltomy and Lalouani, 2024);
- rules distilled from a logical neural network (Xu et al., 2024);
- domain-aware layer-wise relevance propagation (XAI-CPS; Rasheed et al., 2026).

These works mostly validate interpretability qualitatively, on selected attacks, or through expert agreement (Rasheed et al., 2026). They do not report how often the documented attack target is ranked first at the moment of detection.

Post-hoc XAI for CPS, ICS and IIoT intrusion detection. Most of the literature applies SHAP or LIME to a detector and presents feature-importance plots:

- Kernel or Gradient SHAP on autoencoder-based detectors for SCADA and gas-pipeline data (Huong et al., 2022; Ha et al., 2022);
- SHAP in privacy-preserving federated detectors (Yazdinejad et al., 2025);

- correlation-based and unsupervised IIoT detectors (Birihanu and Lendák, 2024; Abudurexiti et al., 2025);
- microgrid attack classifiers (Trivedi et al., 2025);
- process-oriented attacks (Kenmogne and Mocanu, 2024, 2025);
- network-traffic IIoT/IoT detectors (Oseni et al., 2023; Khan et al., 2022; Shoukat et al., 2025; Alani, 2023; Attique et al., 2024; Ahakonye et al., 2024; Saiyed et al., 2026);
- surrogate and example-based explanations (Zolanvari et al., 2023; Panesar et al., 2026).

Among the studies we reviewed, several evaluate on random proportional splits or k-fold cross-validation of time-series data without stating a chronological or gapped protocol (Table B.12). This is a known source of optimistic bias in security ML (Arp et al., 2022; Kaufman et al., 2012).

Evaluating explanations of intrusion detectors. Warnecke et al. (2020) proposed security-specific criteria for explanation methods (accuracy, sparsity, stability, efficiency, robustness, completeness). Wei et al. (2023) and Arreche et al. (2024) evaluate explanations of network IDSs along similar axes, including explanation time. In CPS settings:

- Oinonen and Morsi (2026) compare seven XAI methods for a substation IDS on interpretability, stability, faithfulness and computation time.
- WaXAI (Mathuros et al., 2024) measures the overlap between top-ranked features and attacked SWaT devices.
- Fung et al. (2024) provide the most systematic ground-truth evaluation. Across 156 manipulations on SWaT, WADI and a TEP simulation, they rank the manipulated feature with eight attribution methods and five detectors. They show that attributions at practical detection timings are inaccurate, and that an ensemble of raw-error, saliency and LEMNA (Guo et al., 2018) attributions helps.

Our work starts from their findings. We change the unit of attribution, exploit the temporal structure of incidents, and add the missing dimension of timeliness.

Grouped and temporal Shapley attributions. Owen values (Owen, 1977) extend Shapley values (Shapley, 1953) to a priori unions of players. Sampling estimators make Shapley values tractable for many players (Castro et al., 2009; Štrumbelj and Kononenko, 2014; Lundberg and Lee, 2017). ShaTS (Franco de la Peña et al., 2026) groups time-series features a priori by time step, by sensor, or by SWaT process stage (six groups), with exact or stratified-sampling computation. It reports execution time per window against a one-second budget. Our loop-level game is an instance of a priori grouping in the ShaTS sense. We differ in four ways:

- the groups are *control loops* rather than process stages, the unit used by ICS attack taxonomies and operators;
- the value function uses an incident-local reference;
- we accumulate attributions across an incident;
- we evaluate against documented targets at operational timing and under deadlines.

In a concurrent preprint, Mishra et al. (2026) compute sensor–time attributions with on-manifold conditional counterfactuals for root-cause analysis on SWaT and server data. They aggregate within a window, whereas we aggregate across alarm windows and evaluate timeliness.

Counterfactual explanations for CPS anomaly detection. A complementary line of work explains an anomaly by what a normal input would have looked like. AR-Pro (Ji et al., 2024) generates diffusion-based counterfactual repairs that satisfy formal properties of the detector score and evaluates them on SWaT, WADI and HAI. JoCE (Jeong et al., 2026) searches for sparse joint time–feature edits that remove a time-series anomaly while preserving temporal continuity. Counterfactuals answer a different question from ours (what would have to change, rather than which loop is responsible), require iterative optimisation per explanation, and have not been evaluated against documented attack targets under deadlines. They could be scored with the same deadline-conditioned metrics.

Timeliness in CPS security. Cárdenas et al. (2011) and Urbina et al. (2016) analyse attacks and detection in terms of their physical impact. Puccetti and Ceccarelli (2026) argue that detection latency must be evaluated together with

Table 1: Positioning against the closest prior work ("–": not reported).

	Fung et al. (2024)	ShaTS (Franco de la Peña et al., 2026)	Oinonen and Morsi (2026)	XAI-CPS (Rasheed et al., 2026)	**This work**
Attribution unit	tag	tag / process stage	pixel	tag	control loop (+tag refinement)
Ground-truth localization	yes	case-based	no	expert agreement	yes (2 benchmarks)
At operational timing	yes (fails)	–	n/a	–	primary setting
Across alarm windows	no	no	no	no	yes (TEA)
Explanation cost	–	per window (GPU)	per explanation (GPU)	12.3 ms (GPU)	CPU 1-thread + evaluation count
Physical deadline	no	no	no	no	yes
Statistical tests	ANOVA/ Pearson	–	–	mean±sd	Friedman, Wilcoxon–Holm, bootstrap

false-positive rates. GiBy (Venugopalan and Adepu, 2025) explains detections through per-sensor safety bounds within milliseconds. To our knowledge, no prior work evaluates whether an *explanation* that is *correct* reaches the operator within a process-derived deadline.

3. Problem formulation and threat model

System model. A historian or SCADA feed delivers process-level tags $\boldsymbol{x}_t \in \mathbb{R}^d$ at a fixed sampling period (1 s in both benchmarks). The tags are partitioned into L control loops $\mathcal{G} = \{g_1, \ldots, g_L\}$, where $g_\ell \subseteq \{1, \ldots, d\}$ contains the setpoint, process variable(s), controller output and actuator feedback of loop ℓ. Tags not owned by a controller (e.g. measured disturbances or simulator states) form auxiliary groups. The partition is engineering metadata, available from the piping-and-instrumentation diagram or controller configuration; it is fixed before any test data is seen.

Detector. The detector is a forecaster f that maps a window $W_t = \boldsymbol{x}_{t-w:t-1}$ to $\hat{\boldsymbol{x}}_t = \boldsymbol{x}_{t-1} + f(W_t)$. For a segment $Z = [W_t; \boldsymbol{x}_t] \in \mathbb{R}^{(w+1)\times d}$ the anomaly score is

$$e(Z) = \frac{1}{d}\sum_{j=1}^{d} \psi\left(\frac{x_{t,j} - \hat{x}_{t,j}}{\sigma_j}\right)^2, \qquad \psi(r) = c\tanh(r/c), \tag{1}$$

where σ_j is the residual standard deviation on held-out normal data (floored at 0.05) and ψ is a differentiable soft clip ($c = 20$). An alarm is raised when the

exponentially weighted score $s_t = \alpha e_t + (1-\alpha)s_{t-1}$ exceeds θ, the q-quantile of s on chronologically held-out normal validation data.

Threat model.

- *Access.* The adversary has gained access to the control network (Purdue levels 1–2). Over an interval $[t_{\text{on}}, t_{\text{off}})$ the adversary can: (a) change a setpoint while hiding the change from the HMI; (b) override or offset an actuator command; (c) spoof a sensor value by bias, freezing or replay, as seen by both the controller and the historian; or (d) combine these, e.g. a CV attack masked by PV replay, on one or two loops. This matches the I/O-point attack taxonomy of HAI (Shin et al., 2021).
- *Goal.* Drive the process toward an unsafe state or disrupt production.
- *Knowledge.* In the main experiments the adversary does not know the detector or the explainer. In the decoy experiment (RQ5) the adversary knows that attributions are shown to operators. In addition to the main attack, the adversary then applies a small bias to the PV of an innocent loop to shift blame.
- *Out of scope.* Network-layer attacks without process effect, compromise of the IDS host, and training-data poisoning.
- *Defender's objective.* Deliver the identity of the compromised loop(s), ideally with the manipulated tags, before the process safety time is consumed.

Deadline-conditioned localization. Consider an attack i with onset $t_{\text{on},i}$, target loops $\mathcal{G}_i^\star$ and first alarm $t_{a,i}$ (undefined if missed). An explainer with latency ℓ produces explanations sequentially. It starts at the alarm and computes one explanation every $\Delta_{\text{eff}} = \Delta\lceil \max(\Delta, \ell)/\Delta \rceil$ seconds, where Δ is the nominal refresh period. The j-th explanation uses data up to $t_{a,i} + j\Delta_{\text{eff}}$ and becomes available at $t_{a,i} + j\Delta_{\text{eff}} + \ell$. Let $\boldsymbol{\rho}_i(D) \in \mathbb{R}^L$ be the loop ranking available at $t_{\text{on},i} + D$, i.e. the last explanation delivered by then.

Definition 1 (DCL)**.** *The deadline-conditioned localization at depth k is*

$$\text{DCL@}k(D) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\Big[t_{a,i} \text{ exists}, \ \exists \text{ delivery by } t_{\text{on},i} + D, \\ \mathcal{G}_i^\star \cap \text{top}_k\, \boldsymbol{\rho}_i(D) \neq \emptyset\Big]. \quad (2)$$

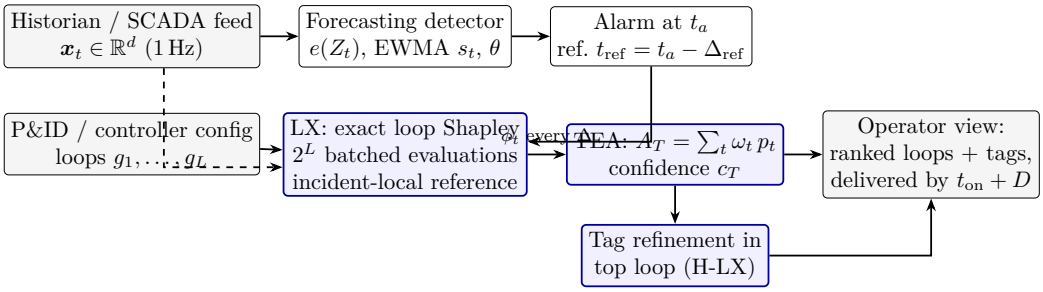


Figure 1: Deadline-aware localization pipeline. Shaded blue: components proposed in this paper. The detector, the explainer and the delivery schedule together determine whether a correct localization reaches the operator before the deadline D.

Missed detections, explanations that are not ready, and wrong explanations all count as failures. Hence DCL@k is upper-bounded by the event recall of the detector.

Definition 2 (In-time rate)**.** *When the hazard deadline D_i^{haz} of each attack is known, the in-time rate is $\frac{1}{N_h} \sum_{i:\, D_i^{\text{haz}} < \infty} \mathbb{1}[\cdot]$ with $D = D_i^{\text{haz}} - t_{\text{act}}$ per attack. Here D_i^{haz} is the time from onset until the* true *physical state first violates a safety limit with no operator response, and t_{act} is a reserve for the operator's action.*

D_i^{haz} is the empirical counterpart of the process safety time of IEC 61511 (International Electrotechnical Commission, 2016). It can be computed in simulation but is not available for recorded datasets. On HAI we therefore report DCL over a grid of deadlines.

4. Method

Figure 1 summarises the pipeline. The detector raises an alarm; from then on, LX explains the anomaly score every Δ seconds at loop level; TEA accumulates these explanations into a ranking with a confidence margin; and the top loop is refined to tags.

4.1. Loop game with an incident-local reference

At alarm time t_a we fix a last-known-good reference time $t_{\text{ref}} = t_a - \Delta_{\text{ref}}$ ($\Delta_{\text{ref}} = 60\,\text{s}$ by default; ablated in Section 6.6). Let $Z_t = \boldsymbol{x}_{t-w:t}$ be the current

segment and $R = \boldsymbol{x}_{t_{\mathrm{ref}}-w:t_{\mathrm{ref}}}$ the reference segment. For a coalition of loops $S \subseteq \mathcal{G}$ define the hybrid segment

$$[Z_t^S]_{\tau j} = \begin{cases} [Z_t]_{\tau j} & j \in \bigcup_{g \in S} g, \\ [R]_{\tau j} & \text{otherwise}, \end{cases} \tag{3}$$

and the characteristic function $v_t(S) = e(Z_t^S) - e(Z_t^\emptyset)$.

The game asks how much of the current anomaly is explained by each loop's trajectory departing from its last known good trajectory, with all other loops restored to that state. This differs from the common interventional value function, which averages over a background drawn from training data (Lundberg and Lee, 2017; Franco de la Peña et al., 2026). Background windows come from unrelated operating points (other setpoints, other plant states). Hybrid segments built from them are off-manifold for a forecaster and can themselves look anomalous. The incident-local reference keeps hybrids close to the operating point of the incident. We compare both choices empirically (LX vs. LX-bg).

*4.2. Exact loop Shapley values (*LX*)*

The attribution of loop g at time t is its Shapley value (Shapley, 1953)

$$\phi_t(g) = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|!\,(L - |S| - 1)!}{L!} \Big[v_t(S \cup \{g\}) - v_t(S) \Big]. \tag{4}$$

Proposition 1. LX *requires exactly* 2^L *evaluations of* $e(\cdot)$ *per explanation, independent of the data and of* d*. It satisfies efficiency,* $\sum_g \phi_t(g) = e(Z_t) - e(R)$*. Since it is exact, it also satisfies the symmetry, dummy and additivity axioms.*

Proof. All 2^L coalitions are evaluated once, and Eq. (4) is a fixed linear map of these values. The axioms are those of the Shapley value (Shapley, 1953), which hold because no sampling approximation is involved. □

The 2^L hybrid segments are evaluated in one batch. For $L \leq 10$ this is at most 1024 forward passes of a small network. Explanation latency is therefore *deterministic and known before deployment*, a property that sampling-based estimators lack. Exact Shapley values over d tags would need 2^d evaluations ($d = 57$ on HAI). Tag-level sampling estimators trade cost for variance (Castro et al., 2009; Štrumbelj and Kononenko, 2014). Grouping by loops is an a

priori union in the sense of Owen (1977), as in ShaTS's multi-feature grouping (Franco de la Peña et al., 2026). The loop is also the unit in which ICS attacks are specified (Shin et al., 2021). Finally, loop grouping encodes the sensor–actuator coupling that Fung et al. (2024) identify as a cause of attribution failure: a loop's SP, PV, CV and feedback enter or leave a coalition together.

*4.3. Temporal evidence accumulation (*TEA*)*

An incident produces a sequence of explanations at times $t_a, t_a + \Delta, \ldots$. Single-shot attributions are noisy. Worse, as the attack propagates through coupled loops, later residuals increasingly appear in loops that merely *react*. Following Fung et al. (2024), we expect the earliest windows to be most informative but not always reliable. TEA therefore aggregates normalised positive attributions, weighted by severity:

$$p_t(g) = \frac{\max(\phi_t(g), 0)}{\sum_h \max(\phi_t(h), 0)}, \qquad A_T(g) = \sum_{t=t_a}^{T} \omega_t \, p_t(g), \qquad \omega_t = \log^+ \frac{s_t}{\theta} + \varepsilon. \tag{5}$$

The ranking published at T sorts loops by A_T, with confidence margin $c_T = (A_T^{(1)} - A_T^{(2)}) / \sum_g A_T(g)$, where $A_T^{(r)}$ is the r-th largest accumulated value. A deployment may publish as soon as $c_T \geq \gamma$, or at the deadline, whichever is first. TEA is agnostic to the attribution method, so we apply it to every baseline as well.

*4.4. Tag refinement (H-*LX*)*

Operators eventually need tags. Within the top-ranked loop $g^\star$ we compute exact Shapley values over its $|g^\star|$ tags. All other loops are held at their current values, so the game isolates within-loop responsibility; the cost is $2^{|g^\star|}$ evaluations ($\leq 2^{12}$ on HAI).

5. Experimental design

5.1. Research questions

- **RQ1 (structure).** Does loop-level exact attribution localize the attacked loop better than tag-level attributions aggregated post hoc? Is any gain due to control structure, or to grouping per se?
- **RQ2 (time).** Does TEA improve accuracy and temporal stability after the alarm?

Algorithm 1 Deadline-aware localization for one incident

Require: detector (f, σ, θ), loops $\mathcal{G}$, alarm time t_a, refresh Δ, deadline D, threshold γ

$t_{\text{ref}} \leftarrow t_a - \Delta_{\text{ref}}$; $R \leftarrow \boldsymbol{x}_{t_{\text{ref}}-w:t_{\text{ref}}}$; $A \leftarrow \mathbf{0} \in \mathbb{R}^L$

for $t = t_a, t_a + \Delta, \ldots$ **while** $t + \ell \leq t_{\text{on}} + D$ **do**

  Build the 2^L hybrid segments Z_t^S; evaluate $e(\cdot)$ in one batch

  $\boldsymbol{\phi}_t \leftarrow$ Eq. (4); $A \leftarrow A + \omega_t \boldsymbol{p}_t$ (Eq. (5))

  Publish ranking argsort$(-A)$ and margin c; **if** $c \geq \gamma$ **then** publish H-LX tags for the top loop

end for

- **RQ3 (deadline).** Which explainers deliver correct localizations before a deadline once latency is accounted for?
- **RQ4 (noise).** How robust are localization and stability to sensor noise?
- **RQ5 (decoy).** How robust are they to a decoy manipulation?
- **RQ6 (rigour).** How much does a random window split inflate detection performance?

Hypotheses and the decision rule (two-sided tests, Holm-corrected $p < 0.05$ with the predicted sign) were fixed in the project's design document before the main runs. They are reproduced in Appendix Appendix A.

5.2. Benchmarks

HAI 21.03. HAI 21.03 (Shin et al., 2021) is a hardware-in-the-loop testbed coupling a boiler (P1), a turbine (P2) and a water-treatment process (P3) with a simulated steam-turbine and pumped-storage power plant (P4). It records 79 process tags at 1 Hz: three normal training files (9.5 days) and five test files with 50 attacks (25 single, 25 combined). Its documentation lists, for each attack, the target controller(s) (P1-PC, P1-LC, P1-FC, P2-SC, P2-TC, P3-LC) and the manipulated points.

- *Ground-truth table.* We transcribed this information into a machine-readable table and matched every entry to the label runs in the data. All 50 start times agree within 2 minutes. Three documented point names differ from the column names in 21.03 and were mapped: `P2_SCO`→`P2_CO_rpm`, `P2_AutoSD`→`P2_ASD`, and the typo

P2_LCV01D→P3_LCV01D. To check the transcription, we compared it with an independent machine-readable transcription of the same manual table, produced separately within the project. The two agree on the target loops and points of 49 of 50 attacks. The one disagreement, A401, is a table row that continues across a page break: the second scenario (AP12 on P1-LC) appears on the next page together with the continuation of the date. We keep both scenarios, as for every other attack of the A4xx series, but excluding A401 changes no reported HR@1 by more than 0.005.

- *Loop map.* Tags were assigned to ten loop groups from the tag descriptions, before any test data was examined. Assignments that the documentation does not make explicit are marked as engineering judgement in the released configuration.

- *Constant tags.* Twenty-two tags are constant throughout normal training and are removed, leaving $d = 57$ tags in $L = 9$ non-empty groups. Manipulated setpoints are largely invisible in the recorded data. Of the 36 (attack, setpoint-tag) target pairs, 30 are exactly constant during the attack, and only 2 vary beyond their range in the preceding 10 minutes (both on P1_B2016). The setpoints of the P2 attacks (P2_ASD, P2_VTR01/02, P2_RTR) never change at all. This is consistent with attacks that hide setpoint changes from the HMI. Tag-level localization of those points is therefore impossible from the data, whereas loop-level localization is not.

PCL-Sim. The simulator models a supply tank, a transfer valve, a heated tank, a centrifugal pump and a discharge valve, under five PI loops:

- LC1: level of tank 1 via the inlet valve;
- FC1: transfer flow via the transfer valve;
- LC2: level of tank 2 via pump speed;
- TC1: temperature of tank 2 via heater power;
- PC1: discharge pressure via the discharge valve.

The plant model includes first-order actuator dynamics, a pump curve, hydraulic and thermal coupling, sensor noise, random-walk disturbances (inlet

temperature, supply pressure) and an operator schedule that changes setpoints every 15–45 min. It exposes $d = 22$ historian tags in $L = 6$ groups (five loops plus a disturbance group).

- *Attacks.* Seven primitives mirror HAI: SP, CV offset, PV bias, PV freeze, CV+PV-freeze, SP+PV-freeze and short CV pulses. They are applied to each loop at two magnitudes and four test seeds, for 280 attack episodes of 1 h with an onset at 25–32 min and a duration of 5–15 min.

- *Hazard deadlines.* Controllers and the historian see the manipulated values, while the true state is integrated separately. The hazard deadline D^{haz} is the time to the first violation of a safety limit (tank overflow or dry-run, temperature $> 85\,^{\circ}\text{C}$, pressure > 8.5 bar). 113 of the 280 attacks are hazardous, with median $D^{\text{haz}} = 301$ s (range 6–1039 s).

- *Normal data.* Twelve 6-hour runs (seeds 0–11, disjoint from the test seeds) are used for training and validation.

5.3. Detectors and leakage control

Four forecaster families are trained on normal data only, each with three seeds, window $w = 32$, and the persistence-plus-correction form of Section 3:

- a linear autoregressive model (4 lags, closed-form ridge);

- an MLP;

- a dilated 1D-CNN (cf. Kravchik and Shabtai, 2018);

- a GRU.

The protocol is:

- Normal files are split chronologically (first 80 % train, last 20 % validation).

- Standardisation statistics come from the training portion. Residual scales σ_j and the alarm threshold θ ($q = 0.999$ on validation EWMA scores, $\alpha = 0.1$) come from validation data only.

- No test labels, test files or test seeds are used for any fitting or hyperparameter choice. All hyperparameters were fixed before the evaluation runs and are listed in the released configuration.
- The linear model is fitted in closed form on all training windows and is therefore identical across seeds.

5.4. Explainers

All explainers attribute the same score $e(Z_t)$ and, unless stated otherwise, use the same reference segment.

- *Tag-level baselines*, summed per loop:
  - raw squared residuals (RE; the "MSE" ranking of Fung et al., 2024);
  - saliency (SM; Simonyan et al., 2014);
  - integrated gradients from the reference (IG, 32 steps; Sundararajan et al., 2017);
  - KernelSHAP (Lundberg and Lee, 2017) with 2^L coalitions (KS-eq, equal budget to LX) or 1024 coalitions (KS-1024);
  - LIME (Ribeiro et al., 2016) with the same two budgets;
  - the ensemble of Fung et al. (2024): normalised RE + SM + a black-box method, with $\beta = 2.5$ for actuators. We use LIME in place of LEMNA (Guo et al., 2018).
- *Structure controls* (exact Shapley over L groups, with group values spread uniformly over member tags):
  - CLX: groups are average-linkage clusters of $1 - |\text{corr}|$, akin to Partition/Owen SHAP;
  - RPX: a random partition with the loop group sizes.
- *Reference control.* LX-bg is LX with an interventional training background of 8 windows, i.e. ShaTS-style multi-feature grouping with loops as groups (Franco de la Peña et al., 2026).
- *ShaTS-sensor.* Feature-grouped stratified-sampling Shapley with $m = 20\,|G|$ coalitions per group, as in Franco de la Peña et al. (2026). Its cost ($\approx 2 \cdot 20\,d^2$ evaluations) restricts it to practical timing on two detectors.

Every method is run single-shot and with TEA.

5.5. Protocol and metrics

Timing. Following Fung et al. (2024), we evaluate at two timings:

- *best-guess*: an oracle window starting at the attack onset, with the reference just before the onset;
- *practical*: from the first alarm, every $\Delta = 5\,\mathrm{s}$ for 180 s; for deadlines beyond this horizon the last explanation is kept.

Metrics.

- *Detection.* Point-wise precision, recall and F1 (no point adjustment; Kim et al., 2022), event recall (alarm within the attack or 60 s after it), median detection delay, and false alarms per hour of attack-free data.
- *Localization.* HR@k (a target loop among the top k) and normalised rank, using the best-ranked target for two-loop attacks.
- *Faithfulness.* Deletion AUC: loops are restored in ranked order, and we take the area under the normalised score curve (lower is better). The restoration uses either the incident reference or the training mean; the latter avoids sharing the value function of LX.
- *Stability.* Top-k agreement between consecutive explanations (temporal), and Jaccard/Spearman agreement under Gaussian input noise ($\sigma = 0.1$ in standardised units).
- *Cost.* Median wall-clock latency on one CPU thread and number of detector evaluations.
- *Timeliness.* DCL@$k(D)$ and the in-time rate with $t_{\mathrm{act}} = 30\,\mathrm{s}$.

Statistics. The unit of analysis is the attack. Seeds are averaged per attack and detector, and (attack, detector) pairs are the paired units. We report means with 95 % bootstrap confidence intervals (10 000 resamples). We test with a Friedman omnibus and two-sided Wilcoxon signed-rank tests of LX+TEA against every other configuration, Holm-corrected (Demšar, 2006; Holm, 1979), with matched-pairs rank-biserial effect sizes r_{rb}.

Implementation. Python 3.13 and PyTorch 2.15 on an Intel Xeon Processor at 2.10GHz. Latencies are measured with `torch.set_num_threads(1)` on an otherwise idle machine; evaluation counts are hardware-independent. We did not measure on embedded hardware (Section 7.1).

Table 2: Detection performance (mean±sd over 3 seeds; no point adjustment). Event recall: alarm within the attack or 60 s after it. FA/h: false-alarm runs per hour of attack-free data.

Data	Detector	P	R	F1	Event rec.	Delay [s]	FA/h
PCL-Sim	Linear	0.79	0.17	0.28	0.96	6	1.38
	MLP	0.81±0.06	0.07±0.04	0.13±0.07	0.76±0.03	226±176	0.31±0.03
	CNN	0.83±0.03	0.19±0.02	0.31±0.02	0.89±0.00	46±38	0.96±0.06
	GRU	0.77±0.02	0.42±0.01	0.54±0.01	0.93±0.00	7±1	0.89±0.05
HAI 21.03	Linear	0.79±0.00	0.60±0.00	0.68±0.00	0.92	6	0.42±0.03
	MLP	0.75±0.01	0.50±0.02	0.60±0.02	0.90	15±1	0.36±0.04
	CNN	0.74±0.01	0.48±0.01	0.58±0.01	0.89±0.01	14±0	0.34±0.01
	GRU	0.78±0.00	0.55±0.00	0.65±0.00	0.90	14±1	0.36±0.02

6. Results

6.1. Detection

Table 2 gives detection performance.

- *HAI.* All four detectors detect 89–92 % of the 50 attacks at 0.34–0.42 false alarms per hour. The median delay is 6 s (linear) to 15 s.
- *PCL-Sim.* Event recall is 76–96 %. Point-wise recall is low because the PI controllers partly compensate the manipulations, and several attacks become visible only through slow drifts.
- *Early alarms.* In ten HAI attacks the score crosses θ at or just before the labelled onset. In nine of them the score rises 1–4 s before the label, which suggests that the documented onsets lag the manipulations slightly. In A301 the score already exceeds θ several seconds before the labelled onset, so an alarm run that began before the attack is counted as a detection at onset. We keep the documented labels throughout.

Localization is evaluated on detected attacks: 181 (attack, detector) units on HAI and 1014 on PCL-Sim. Deadline metrics count missed detections as failures (200 and 1120 units).

6.2. RQ1: Control-loop structure

Tables 3 and 4 report loop localization for every method.

- *At the alarm, HAI.* LX ranks the attacked loop first in 0.79 of attacks, the highest of all methods. The differences are significant against KernelSHAP-1024, equal-budget KernelSHAP and LIME, LX-bg, saliency and both structure controls. Against IG, LIME-1024, the Fung ensemble and RE they are borderline (Holm p between 0.05 and 0.06; Table 5).

- *At the alarm, PCL-Sim.* LX reaches 0.85, within two points of the best (LX-bg, the Fung ensemble). Most methods perform similarly here, because the simulated attacks are localized early.

- *H1 is therefore only partially supported.* Exact loop attribution is the most accurate *black-box* method at the alarm on HAI. It is not uniformly better than white-box integrated gradients, and on PCL-Sim it is not better than the training-background variant.

For reference, a uniformly random ranking of the groups would place an attacked loop first in 0.17 of the same HAI units (multi-loop attacks count as hits on any target loop) and in 0.17 on PCL-Sim; within the top three, in 0.45 and 0.50. All attribution methods are well above chance. The structure controls come closest to it (random partition 0.25 on HAI, correlation clusters 0.27 on PCL-Sim).

Structure, not grouping, is what matters (H4). Exact Shapley values over the same number of groups formed by correlation clustering (CLX) or at random (RPX) are far worse. On HAI at the alarm they reach 0.45 and 0.25 against 0.79; on PCL-Sim, 0.27 and 0.54 against 0.85 (all Holm $p < 10^{-10}$). Data-driven groups mix tags of different loops, so a group can be "responsible" without identifying a loop. Grouping by control structure is what turns exact Shapley values into an operational localization. H4 is supported on both benchmarks.

Best-guess timing. Computing attributions on an oracle window that starts at the onset, as advocated by Fung et al. (2024), does *not* help in our setting. On HAI it is worse than practical timing for every method except saliency (e.g. LX 0.47 vs. 0.79), because many HAI and PCL-Sim manipulations take tens of seconds to affect the process. Their detectors mostly fire after the effect has become visible.

Table 3: Loop localization on HAI 21.03 (pooled over 4 detectors; mean over attacks, seeds averaged). Bold: best in column. Single-shot unless marked TEA.

	HR@1						HR@3	nRank
Method	best-guess	alarm	+60 s	+60 s TEA	+180 s	+180 s TEA	+60 s TEA	+60 s TEA
RE	0.49	0.71	0.48	0.61	**0.35**	0.57	0.89	0.09
SM	0.43	0.43	0.32	0.44	0.22	0.44	0.67	0.27
IG	0.50	0.75	0.56	**0.72**	0.33	0.70	0.93	**0.06**
KernelSHAP-eq	0.47	0.70	0.50	0.64	0.22	0.60	0.90	0.09
KernelSHAP-1024	0.48	0.74	0.51	0.69	0.28	0.65	0.92	0.07
LIME-eq	0.49	0.75	0.55	0.70	0.26	0.66	0.93	0.07
LIME-1024	0.49	0.76	0.56	0.69	0.31	0.66	0.93	0.07
Fung ens.	**0.60**	0.76	0.56	**0.72**	0.34	**0.71**	0.92	0.07
CLX (clusters)	0.31	0.45	0.39	0.38	0.20	0.38	0.79	0.18
RPX (random)	0.17	0.25	0.19	0.21	0.16	0.17	0.68	0.26
LX-bg	0.55	0.74	0.58	0.71	0.31	0.65	**0.94**	**0.06**
LX (ours)	0.47	**0.79**	**0.60**	0.70	0.30	0.64	**0.94**	**0.06**

Hard-to-explain attacks. Table D.16 stratifies the HAI results by the type of manipulated variable, read from the target-controller code of the official attack table (e.g. `P1-PC-SP1PV1`: setpoint plus a spoofed process variable). The strata are small (3–23 attacks), so we describe them only.

- *Controller-output (CV) attacks* directly move a recorded actuator and are localized well (at the alarm RE 0.69; IG, KernelSHAP-1024, LX-bg and LX 0.79; LX 1.00 with a PV spoof, $n = 3$).
- *Setpoint (SP) attacks* are the hardest. The manipulated setpoint itself is invisible in the data (Section 5), so the evidence appears only in the loop's PV and CV. LX is the best method at the alarm (0.73 vs. 0.66 for IG). However, every method falls to 0.38–0.52 at 60 s even with TEA, plausibly because the controller has by then propagated the change to coupled loops.

This pattern supports loop-level attribution for CPS more generally. Only the loop is identifiable when the manipulated variable is not observable; any tag-level ground truth that names a hidden setpoint cannot be met by any explainer.

6.3. RQ2: Attribution after the alarm and temporal evidence accumulation

Figure 2 shows the central phenomenon. Single-shot attributions degrade quickly after the alarm, for *every* method:

- on HAI, LX falls from 0.79 at the alarm to 0.60 after 60 s and 0.30 after 180 s;

Table 4: Loop localization on PCL-Sim (pooled over 4 detectors; mean over attacks, seeds averaged). Bold: best in column. Single-shot unless marked TEA.

	HR@1						HR@3	nRank
Method	best-guess	alarm	+60 s	+60 s TEA	+180 s	+180 s TEA	+60 s TEA	+60 s TEA
RE	0.43	0.86	0.44	0.80	0.41	0.80	0.96	0.07
SM	0.44	0.65	0.43	0.70	0.40	0.69	0.95	0.10
IG	**0.68**	0.86	**0.66**	0.84	**0.55**	**0.83**	0.96	**0.06**
KernelSHAP-eq	0.46	0.80	0.47	0.71	0.40	0.69	0.90	0.13
KernelSHAP-1024	0.50	0.84	0.50	0.80	0.43	0.77	0.93	0.09
LIME-eq	0.42	0.82	0.46	0.72	0.40	0.70	0.91	0.12
LIME-1024	0.50	0.85	0.52	0.82	0.45	0.81	0.95	0.07
Fung ens.	0.48	**0.87**	0.49	0.82	0.45	0.82	**0.97**	**0.06**
CLX (clusters)	0.12	0.27	0.16	0.23	0.14	0.23	0.78	0.31
RPX (random)	0.35	0.54	0.35	0.58	0.32	0.58	0.90	0.16
LX-bg	0.54	**0.87**	0.60	**0.85**	0.50	**0.83**	0.96	**0.06**
LX (ours)	0.57	0.85	0.53	0.82	0.46	0.80	0.95	0.07

- on PCL-Sim, it falls from 0.85 to 0.53 and 0.46.

This is the closed-loop propagation mechanism. Once controllers react, the largest residuals move to loops that compensate for the manipulated one. The mechanism complements the timing effect reported by Fung et al. (2024): attributions are best near the first visible effect and degrade afterwards.

TEA *recovers most of this loss (H2).*

- LX. With TEA, LX reaches 0.70 (HAI) and 0.82 (PCL-Sim) at 60 s, and 0.64 and 0.80 at 180 s. Against single-shot LX at 60 s: Holm $p = 0.010$ with $r_{\text{rb}} = 0.48$ on HAI, and $p = 6.7e - 56$ with $r_{\text{rb}} = 0.83$ on PCL-Sim.

- *Other methods.* The gain holds for all methods; e.g. RE on HAI improves from 0.48 to 0.61 at 60 s.

- *Shortlists.* HR@3 with TEA is 0.94 (HAI) and 0.95 (PCL-Sim): the attacked loop is almost always among three candidates. This matches the operator preference for short candidate lists reported by Fung et al. (2024).

- *Temporal stability* (Fig. 5, Table 9). Single-shot top-1 loops change between consecutive explanations 5 s apart in 23–48 % of cases, depending on the method and benchmark. With TEA, consecutive explanations agree in 95–99 % of cases for every method.

H2 is supported.

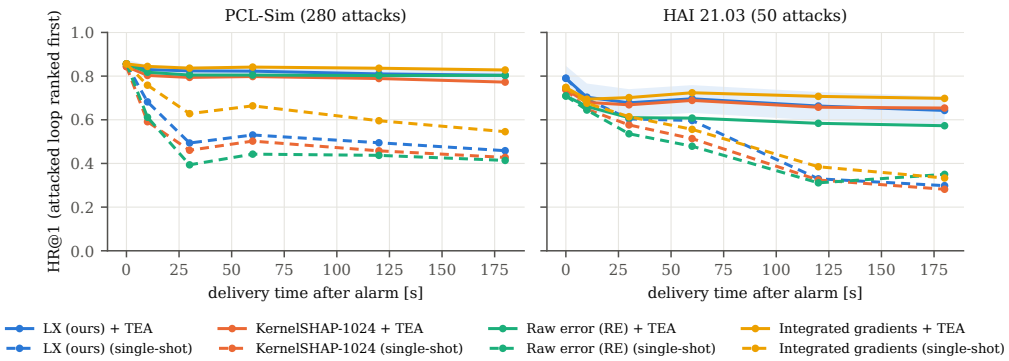


Figure 2: Top-1 loop localization as a function of the delivery time after the alarm (solid: with TEA; dashed: single-shot; band: 95 % bootstrap CI for LX+TEA). Every single-shot method decays as the attack propagates through coupled loops; TEA retains most of the accuracy reached at the alarm.

With TEA*, the leading methods are statistically tied.* At 60 s on HAI, LX+TEA (0.70) is not significantly different from IG, the Fung ensemble, LX-bg, LIME-1024 or KernelSHAP-1024, each with TEA (Table 5). It is significantly better than RE, equal-budget KernelSHAP, saliency, CLX and RPX. On PCL-Sim, IG+TEA (0.84) and LX-bg+TEA (0.85) are slightly but significantly better than LX+TEA (0.82).

6.4. RQ3: Deadlines, latency and cost

Tables 6 and 7 and Fig. 3 report deadline-conditioned localization. Detection delay, explainer latency and the correctness of the delivered explanation are all accounted for.

- *Ceiling.* Because DCL counts missed and late detections as failures, it is bounded by event recall. On HAI, DCL@1 at $D = 60$ s is 0.53 for LX+TEA, and 0.60 at 120 s. DCL@3 is 0.70 at 60 s.
- *No winner on HAI.* None of the leading methods is significantly better than the others at 60 s (Table 5).
- *Hazardous attacks on PCL-Sim.* The correct loop is ranked first before the hazard deadline minus a 30 s action reserve in 0.48 of cases with TEA, versus 0.28 single-shot. With a three-loop shortlist the rate is 0.53.

- *What limits the in-time rate.* The remaining failures are dominated by detection, not explanation. Of the 704 failing (attack, detector, seed) cases of LX+TEA, 603 (86 %) are attacks that were missed or detected too late for any explanation to be delivered. Conditional on delivery, the top-ranked loop is correct in 0.87 of cases. The corresponding figures are 0.90 for IG, 0.89 for LX-bg, 0.74 for RE and 0.78 for KernelSHAP-1024.

Latency and determinism. On one CPU thread, LX costs 2^L evaluations per explanation: 512 on HAI, 64 on PCL-Sim (Table 8). Across the four detectors this takes 0.3–2.5 ms on PCL-Sim and 2.2–16.9 ms on HAI, far below the 1 s sampling period, so latency barely affects DCL at these speeds. Two properties distinguish LX:

- *Determinism.* Its cost is fixed and known before deployment. Equal-budget sampling estimators (KernelSHAP-eq, LIME-eq) are less accurate at the same cost (Tables 3 and 4). Approaching LX's accuracy with them requires a larger budget (KernelSHAP-1024, LIME-1024: twice the evaluations on HAI, sixteen times on PCL-Sim).

- *Gradient-free.* IG is the cheapest accurate method (32 forward/backward passes), but it requires gradient access to the detector. LX needs only black-box score queries, so it also applies to non-differentiable detectors (rules, trees, PASAD-like statistics).

Sensitivity to slower hardware. We did not measure on embedded hardware. As a crude proxy, we repeated the deadline analysis with every measured latency multiplied by an assumed factor of 10 or 100; the explanation refresh interval grows accordingly. For LX+TEA on HAI, DCL@1 at $D = 30$ s is 0.51 at the measured latency, 0.51 at $\times 10$ and 0.50 at $\times 100$. On PCL-Sim, the in-time rate is 0.48, 0.48 and 0.48. Only LX-bg on HAI, with 4096 evaluations per explanation, is materially affected: at $D = 30$ s it drops from 0.51 to 0.28 at $\times 100$. All other configurations change by at most 0.04, in either direction (largest for the 1024-sample estimators on HAI). Changes can be positive because a slower explainer refreshes less often, and TEA then accumulates a different subset of windows. At these model sizes, explanation latency is not the binding constraint even with a hundredfold slowdown; detection delay is.

ShaTS-sensor at its published default budget needs $\approx 2 \cdot 20\, d^2$ evaluations per explanation (129960 on HAI), 4.8 s for the CNN on one thread. At the alarm on HAI it localizes the attacked loop in 0.67 of attacks, against 0.73

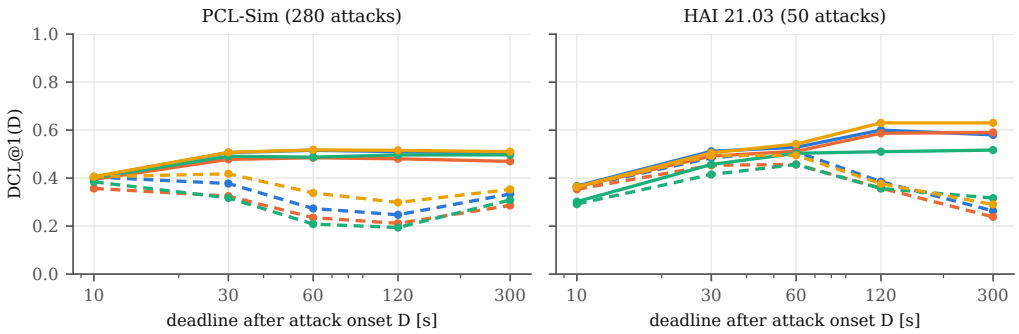


Figure 3: Deadline-conditioned localization DCL@1(D) (colours as in Fig. 2; solid: TEA; dashed: single-shot). Missed detections and explanations not yet delivered count as failures, so curves are bounded by event recall.

for LX on the same incidents (CNN, seed 0, $n = 45$); for the linear detector the figures are 0.83 and 0.85 ($n = 46$). On PCL-Sim (19360 evaluations, 0.6 s for the CNN) both reach 0.80 (CNN) and 0.89 (linear). On these subsets ShaTS-sensor is at most on par with loop-level attribution (no significance test; small n on HAI) while taking roughly 250–280 times longer for the CNN; at ≈5 s per explanation on HAI it could not be refreshed every second.

6.5. Explanation quality beyond accuracy

- *Faithfulness.* Deletion AUCs at the alarm (restoration to the training mean; Table 9) are similar for all accurate methods (0.16–0.21). Saliency and the structure controls are clearly worse. Deletion with the incident reference was numerically unstable (the normaliser $e(Z) - e(R)$ can approach zero), so we do not report it.

- *Stability under input noise.* LX is *less* stable than the sampling estimators: top-3 Jaccard 0.56 vs. 0.70 for KernelSHAP-1024 on HAI. Exact attributions follow every change of the input, whereas the fixed-seed sampling designs smooth small differences. TEA mitigates this in operation, because individual noisy explanations carry limited weight.

- *Tag-level localization.* Within the top loop, H-LX ranks a manipulated tag first in 0.31 (HAI) and 0.45 (PCL-Sim) of attacks, and among the top three in 0.49 and 0.76. This is comparable to the best tag-level methods; the Fung ensemble is best on HAI for tag HR@1.

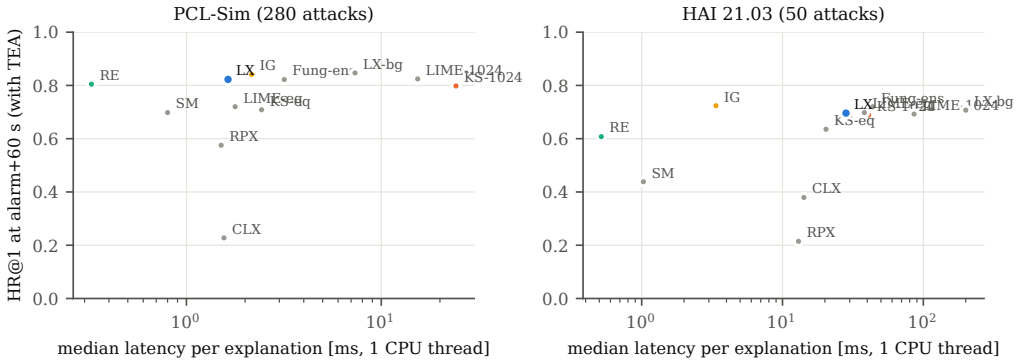


Figure 4: Accuracy at 60 s after the alarm (with TEA) versus median latency per explanation (1 CPU thread, measured during the runs).

6.6. Ablations and robustness (RQ4, RQ5)

Table 10 summarises the ablations. The reference-gap and noise conditions use one detector (CNN, seed 0), so they have less statistical power than the main comparison ($n \approx 45$ attacks on HAI).

Reference gap Δ_{ref}. The incident-local reference is taken Δ_{ref} seconds before the alarm.

- *PCL-Sim.* LX is insensitive to the gap. At the alarm it reaches 0.79, 0.80, 0.80 and 0.81 for 32, 60, 120 and 300 s. With TEA at 60 s it reaches 0.76–0.78.
- *HAI.* The default of 60 s is the best of the four gaps at the alarm (0.73). A 32 s gap gives 0.67 and a 300 s gap 0.64. With TEA at 60 s the range is 0.62–0.67.

A short gap risks a reference that already contains the start of the manipulation. A long gap risks one from a different operating point. On HAI the differences correspond to a few attacks out of about 45, and we did not tune the gap on test data. As expected, RE and LX-bg do not depend on the reference and are unchanged. IG and KernelSHAP use the same reference as their baseline and vary similarly to LX.

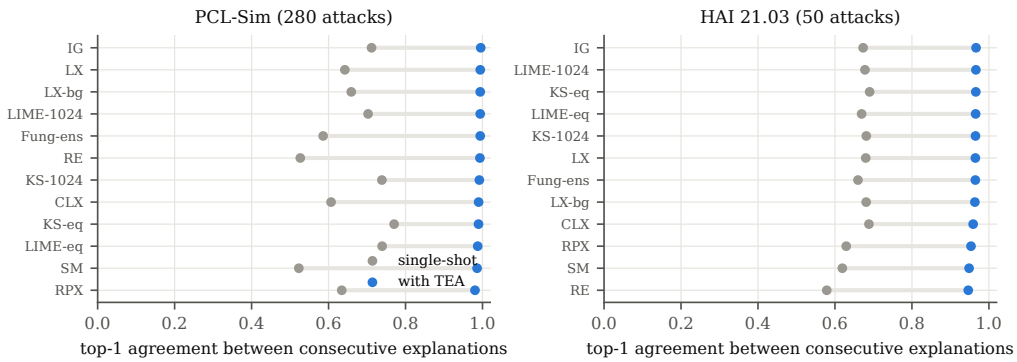


Figure 5: Temporal stability: share of consecutive explanations (5 s apart, first 120 s after the alarm) with the same top-1 loop, single-shot (grey) vs. with TEA (blue).

Sensor noise (RQ4). We add i.i.d. Gaussian noise with standard deviation $\sigma_n \in \{0.1, 0.3\}$ (in standardised units) to the test data.

- *Clean threshold.* If the residual scale σ and the threshold θ are kept from clean validation data, the detector alarms almost continuously. Precision falls to 0.02 on HAI and 0.17 on PCL-Sim, and every attack is "detected" by an alarm that is already running at its onset. Localization numbers in this setting measure a saturated detector rather than the explainer, so we do not interpret them. They are included in the released `phase2.csv`.

- *Recalibrated threshold.* We therefore also recalibrate σ and θ on validation data with the same noise level, keeping the model weights frozen, as an operator would after commissioning noisier sensors.

With recalibration, the two benchmarks respond very differently.

- *HAI.* Event recall and delay are nearly unchanged: event recall 0.88 and 0.86, against 0.90 without noise, with median delays of 20 and 15 s against 14 s. Localization at the alarm degrades for all methods. LX reaches 0.64 and 0.51 against 0.73 without noise; IG reaches 0.61 and 0.56. At 60 s with TEA, LX reaches 0.61 and 0.56. Against IG and KernelSHAP-1024 with TEA, LX is numerically lower but not significantly different (Holm $p \geq 0.61$, $n = 44$ and 43).

- *PCL-Sim.* Noise mainly delays detection. The median delay grows from 18 s to 382 s and 301 s, presumably because slowly developing manipulations must grow larger before they exceed the wider noise floor. By then they are easy to localize: all methods reach 0.91–0.95 at 60 s with TEA, and LX+TEA does not differ significantly from any of them (Holm $p \geq 0.08$, $n = 244$ and 247). The cost appears in the deadline: a localization delivered minutes later is of little use against fast hazards.

H5a is therefore *not supported*. LX+TEA does not degrade less than the baselines under sensor noise. Together with the lower perturbation stability of single LX explanations (Table 9), this suggests that exact loop attributions inherit the noise sensitivity of the detector score. Temporal accumulation still helps under noise (e.g. HAI, $\sigma_n = 0.1$: LX 0.48 single-shot vs. 0.61 with TEA at 60 s), but not more for LX than for other methods. Under realistic noise the dominant effect is on detection timeliness, consistent with the deadline decomposition in Section 6.

Decoy manipulation (RQ5). In the decoy suite, every attack is accompanied by a small bias on the PV of an innocent loop. The decoy rarely captures the top rank for any method:

- at the alarm, 0.03 of attacks for LX and 0.04 for RE;
- at 60 s with TEA, 0.05 for LX and 0.04–0.05 for the other methods.

Single-shot explanations 60 s after the alarm are the most vulnerable. Single-shot LX then ranks the decoy first in 0.14 of attacks, plausibly because the controllers have partly compensated for the main attack by then, so that the decoy residual is relatively larger. None of the paired differences in decoy capture between LX+TEA and the other methods with TEA is significant (all Holm $p \geq 0.23$, $n = 520$ (attack, detector) units). Top-1 accuracy with the decoy is close to that without it (Table 10). H5b is therefore *not supported*: a non-adaptive decoy of this size is ineffective against all methods, so it cannot discriminate between them. An adaptive adversary remains untested (Section 7.1).

6.7. RQ6: Leakage from random window splits

We trained a gradient-boosting classifier on window features (last value, mean and standard deviation of every tag) from the HAI attack files, as supervised XAI-IDS studies do.

- *Random 70/30 split.* F1 $= 0.940 \pm 0.008$ and average precision 0.983 (5 seeds).
- *Chronological, attack-disjoint split* (train on test1–3, evaluate on test4–5). The same model reaches F1 $= 0.575 \pm 0.029$ and AP 0.624.

A random split inflates F1 by ≈0.37 because overlapping windows of the same attack instance fall on both sides of the split. H6 is supported. Explanations computed for models evaluated this way inherit the optimism.

6.8. Summary of hypotheses

7. Discussion

What makes CPS explanations timely?. Our results decompose the timeliness problem into three factors.

1. *Detection dominates.* On PCL-Sim, 86 % of the deadline failures of the best configurations are attacks detected too late or not at all. No explanation method can fix those, so the deadline perspective puts detection latency (Puccetti and Ceccarelli, 2026) back at the centre.
2. *When attributions are computed matters more than how.* Every attribution method we tested, gradient-based, sampling-based or exact, degrades as an attack propagates through coupled loops. Explanations computed a minute or more after the alarm are therefore often misleading if read in isolation. This is consistent with, and adds a mechanism to, the timing failure reported by Fung et al. (2024).
3. *Temporal evidence accumulation is a cheap and general remedy.* It adds no detector evaluations, and it applies unchanged to any attribution method. In our experiments it was the single largest improvement for every method. It also turns flickering top-1 outputs into stable ones.

What does control-loop structure buy?. The structure controls answer a question left open by grouped Shapley methods (Owen, 1977; Franco de la Peña et al., 2026): grouping is only useful if the groups match how the plant is controlled. Correlation clusters and random partitions with the same group sizes perform dramatically worse than control loops.

Given loop structure, exact Shapley values (LX) are the most accurate black-box attribution at the alarm on HAI, and tie with the best methods once TEA is applied. Their operational advantage is predictability, not a large accuracy margin:

- a fixed number of detector queries (2^L), decided at design time and independent of the data;
- no randomness, so the same alarm yields the same explanation;
- no gradient access.

Where gradients are available, integrated gradients aggregated per loop with TEA is an equally strong and cheaper choice in our data. We recommend reporting it as a baseline in future work. The training-background variant (LX-bg) was slightly better on the simulator and slightly worse at the alarm on HAI. We therefore do not claim a general advantage for the incident-local reference. Nor did LX+TEA show the hypothesised robustness advantage: under recalibrated sensor noise and a non-adaptive decoy it behaved like the baselines (Section 6.6). Under noise, the larger effect was on detection timeliness, which again points to detection as the bottleneck.

Practical recommendations.

1. Evaluate explanations at operational timing, against documented targets, and over time, not only at a single window.
2. Group attributions by control loop, using the P&ID or controller configuration.
3. Accumulate evidence across the incident and present a short ranked list (the top three contained the attacked loop in 94–95 % of detected attacks at 60 s in both benchmarks).
4. Report deadline-conditioned metrics alongside latency, because fast explanations of late or wrong alarms do not help the operator.
5. Use chronological, attack-disjoint evaluation. On HAI, a random window split inflated F1 from 0.575 to 0.940.

7.1. Limitations and threats to validity

External validity.

- *Benchmark coverage.* We evaluate on one real HIL testbed (HAI 21.03, 50 attacks) and one simulator of our own design. SWaT and WADI, the most common ICS benchmarks, are only available on request, and we could not include them in this study. Replication on them, on HAI 22.04/23.05, and on TEP-based manipulation suites (Fung et al., 2024) is the most important next step.

- *Other CPS domains.* Both benchmarks are industrial process CPS. Power grids, vehicles and medical devices also have control loops and physical deadlines, so the framework applies to them in principle. Their loop sizes, sampling rates and deadlines differ, however, and we make no empirical claim about them.

- *Simulator bias.* A simulator designed by the authors of a method risks favouring it. We mitigate this in three ways: all explainers query the same score function and reference; attack primitives mirror HAI's taxonomy; and simulator parameters were fixed on development seeds before any explainer was evaluated. The simulator is released for scrutiny.

Deadlines. Hazard deadlines are computed in simulation, with no operator or safety-system response, and with safety limits we chose. On HAI we can only report DCL over a deadline grid, because recorded data carry no process safety times. Real process safety times come from hazard analyses (IEC 61511) and differ by orders of magnitude between processes.

Hardware. Latencies were measured on one thread of a server CPU, not on a PLC-adjacent embedded device. We report hardware-independent evaluation counts and a sensitivity analysis with an assumed slowdown factor, but we did not measure on embedded hardware. Edge-class conclusions are therefore extrapolations.

Method assumptions.

- *Engineering metadata.* LX needs a loop map. Some HAI assignments are our engineering judgement (flagged in the released configuration).

- *Scaling.* Exact computation is exponential in L. Beyond $L \approx 14$ (16k evaluations), loops would need a two-level hierarchy (unit $\rightarrow$ loop) or sampling, and the deterministic-latency guarantee would weaken.

- *Reference.* The incident-local reference assumes that the plant was in a normal state Δ_{ref} seconds before the alarm. Slow ramps that begin long before detection contaminate it (Section 6.6).

- *Detector family.* We study forecasting detectors only. Reconstruction-, density- or classifier-based detectors would need their own score functions.

Adversary. The decoy experiment considers a non-adaptive adversary with a fixed small bias. Explanation-manipulation attacks that preserve the detector's decision while steering SHAP, LIME or integrated-gradients explanations have been demonstrated on security classifiers, including intrusion detection (Slack et al., 2020; Mia and Pritom, 2025). Whether such an adversary can steer a loop-level attribution while respecting the physical constraints of a closed control loop remains open and is the most important robustness question for this work.

Human factors. We measure whether the correct loop is ranked first or within the top k. We do not measure whether operators act faster or better. The preference for showing several candidates found in the operator survey of Fung et al. (2024) motivates our HR@3 results but does not substitute for a user study.

Literature search. Bibliographic APIs (OpenAlex, Crossref, arXiv, Semantic Scholar) were not reachable from our analysis environment. The novelty check therefore combined the curated corpus with general web search. We found no prior work combining the elements above, but this does not prove that none exists.

8. Conclusion

We asked whether the explanations of ML-based intrusion detectors in cyber-physical systems reach operators correctly and in time, and argued that XAI for CPS should be grounded in the control structure and the physical deadlines of the system. We introduced deadline-conditioned evaluation metrics, loop-level exact Shapley attribution with deterministic cost, and temporal evidence accumulation. We evaluated them, with 12 baselines and controls, across four detector families on the public HAI 21.03 testbed, for which we curated loop-level ground truth, and on an open closed-loop simulator with physical hazard deadlines.

Five findings stand out:

- Single-shot attributions degrade quickly after an alarm as attacks propagate through coupled loops, and temporal accumulation recovers most of this loss for every method.
- Attribution grouped by control loops is essential; data-driven groupings fail.

- Most deadline failures are detection failures, not explanation failures.
- Setpoint attacks are the hardest to explain: the manipulated setpoint is invisible in the recorded data, so only the loop, not the tag, can be identified.
- Exact loop attribution (LX) is the most accurate black-box method at the alarm on real testbed data, but it is not more accurate or more robust than the strongest baselines once evidence is accumulated. Its advantages are a predictable, data-independent cost and gradient-free determinism.

Future work should:

- replicate the study on SWaT, WADI and newer HAI releases;
- extend it to other CPS domains, such as power grids and vehicles, where control loops and deadlines can be defined analogously;
- measure latency on embedded hardware;
- study adaptive adversaries who steer the explanation while keeping the alarm, under the physical constraints of a closed loop;
- evaluate deadline-aware explanations with operators.

Data and code availability

All code, configuration files, the HAI 21.03 loop-level ground-truth table, the PCL-Sim simulator, raw per-attack outputs and the scripts that generate every table and figure are available in the project repository (`https://github.com/mazharulmd/XAI-CPS`). HAI 21.03 is publicly available from its maintainers (Shin et al., 2021).

Declaration of generative AI use

[To be completed by the authors according to the journal policy.]

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Appendix A. Pre-specified hypotheses

The following were recorded in `docs/02_research_design.md` before the evaluation runs.

- **H1.** LX achieves higher HR@1 and a lower normalised rank than tag-level attributions aggregated post hoc.
- **H4.** LX beats exact Shapley over data-driven clusters (CLX) and random partitions (RPX) with the same number of groups.
- **H2.** TEA increases HR@k at a fixed delivery time and increases temporal stability.
- **H3.** LX+TEA has deterministic cost, lower latency than sampling explainers at equal or higher accuracy, and the highest DCL and in-time rate.
- **H5a.** Localization degrades less under sensor noise with LX+TEA.
- **H5b.** A decoy captures top-1 less often with LX+TEA.
- **H6.** Random window splits inflate detection metrics relative to chronological, attack-disjoint splits.

A hypothesis is reported as supported only if the Holm-corrected two-sided $p < 0.05$ and the effect has the predicted sign.

Appendix B. Validation protocols in the reviewed corpus

Appendix C. HAI 21.03 loop map

The ten groups used for HAI 21.03 are listed below. Tags constant in training were removed, leaving 57 tags in nine non-empty groups; P1-AUX holds discrete pump-state tags, all constant. The full mapping, with engineering-judgement flags, is in `configs/hai/hai2103_loops.yaml`, and the ground truth is in `configs/hai/hai2103_attacks.csv`.

- P1-PC: pressure;
- P1-LC: return-tank level;
- P1-FC: discharge flow;
- P1-TC: heat-exchanger temperature;
- P1-AUX: pumps;
- P2-SC: turbine speed;
- P2-TC: vibration and trip limits;
- P3-LC: upper-tank level;
- P4-ST: steam-turbine model;
- P4-HT: hydro-turbine model.

Appendix D. Additional results

Table 5: Paired comparisons of LX+TEA against selected configurations (two-sided Wilcoxon signed-rank on (attack, detector) units, Holm correction over all 23 comparisons per row group; $r_{\mathrm{rb}} > 0$ favours LX+TEA). At the alarm TEA has no history, so "LX+TEA" equals LX single-shot.

Data	Metric	Comparator	LX+TEA	Comp.	p_{Holm}	r_{rb}
PCL-Sim	HR@1, alarm	CLX (clusters) + TEA	0.854	0.271	<0.001	+1.00
		Fung ens. + TEA	0.854	0.866	0.101	-0.37
		IG + TEA	0.854	0.856	1.000	-0.20
		KernelSHAP-1024 + TEA	0.854	0.844	0.002	+0.74
		LIME-1024 + TEA	0.854	0.854	1.000	+0.25
		LX single-shot	0.854	0.854	1.000	+0.00
		LX-bg + TEA	0.854	0.870	<0.001	-0.77
		RE + TEA	0.854	0.857	1.000	-0.05
		RPX (random) + TEA	0.854	0.542	<0.001	+0.95
PCL-Sim	HR@1, +60 s	CLX (clusters) + TEA	0.823	0.228	<0.001	+0.96
		Fung ens. + TEA	0.823	0.822	1.000	+0.05
		IG + TEA	0.823	0.841	<0.001	-0.69
		KernelSHAP-1024 + TEA	0.823	0.798	<0.001	+0.45
		LIME-1024 + TEA	0.823	0.824	1.000	+0.01
		LX single-shot	0.823	0.530	<0.001	+0.83
		LX-bg + TEA	0.823	0.847	<0.001	-0.48
		RE + TEA	0.823	0.805	0.062	+0.20
		RPX (random) + TEA	0.823	0.575	<0.001	+0.92
PCL-Sim	DCL@1(60 s)	CLX (clusters) + TEA	0.516	0.096	<0.001	+1.00
		Fung ens. + TEA	0.516	0.502	<0.001	+0.56
		IG + TEA	0.516	0.518	0.462	-0.33
		KernelSHAP-1024 + TEA	0.516	0.485	<0.001	+0.87
		LIME-1024 + TEA	0.516	0.504	0.001	+0.65
		LX single-shot	0.516	0.273	<0.001	+0.98
		LX-bg + TEA	0.516	0.516	1.000	+0.00
		RE + TEA	0.516	0.487	<0.001	+0.77
		RPX (random) + TEA	0.516	0.332	<0.001	+0.99
PCL-Sim	In-time@1	CLX (clusters) + TEA	0.481	0.144	<0.001	+1.00
		Fung ens. + TEA	0.481	0.442	<0.001	+0.74
		IG + TEA	0.481	0.499	0.011	-0.66
		KernelSHAP-1024 + TEA	0.481	0.434	<0.001	+0.83
		LIME-1024 + TEA	0.481	0.465	0.015	+0.54
		LX single-shot	0.481	0.284	<0.001	+0.91
		LX-bg + TEA	0.481	0.494	0.015	-0.67
		RE + TEA	0.481	0.409	<0.001	+0.83
		RPX (random) + TEA	0.481	0.291	<0.001	+0.99
HAI 21.03	HR@1, alarm	CLX (clusters) + TEA	0.790	0.448	<0.001	+0.86
		Fung ens. + TEA	0.790	0.757	0.060	+0.58
		IG + TEA	0.790	0.748	0.052	+0.78
		KernelSHAP-1024 + TEA	0.790	0.738	0.004	+0.87
		LIME-1024 + TEA	0.790	0.755	0.050	+0.81
		LX single-shot	0.790	0.790	1.000	+0.00
		LX-bg + TEA	0.790	0.740	0.012	+1.00
		RE + TEA	0.790	0.709	0.060	+0.46
		RPX (random) + TEA	0.790	0.248	<0.001	+1.00
HAI 21.03	HR@1, +60 s	CLX (clusters) + TEA	0.696	0.379	<0.001	+0.84
		Fung ens. + TEA	0.696	0.720	0.600	-0.37
		IG + TEA	0.696	0.724	0.117	-0.61
		KernelSHAP-1024 + TEA	0.696	0.689	0.600	+0.30
		LIME-1024 + TEA	0.696	0.692	0.684	+0.38
		LX single-shot	0.696	0.597	0.010	+0.48
		LX-bg + TEA	0.696	0.707	0.691	-0.24
		RE + TEA	0.696	0.608	0.004	+0.51
		RPX (random) + TEA	0.696	0.215	<0.001	+0.95
HAI 21.03	DCL@1(60 s)	CLX (clusters) + TEA	0.528	0.322	<0.001	+0.82
		Fung ens. + TEA	0.528	0.540	1.000	-0.29
		IG + TEA	0.528	0.542	1.000	-0.45
		KernelSHAP-1024 + TEA	0.528	0.512	0.578	+0.45
		LIME-1024 + TEA	0.528	0.522	1.000	+0.33
		LX single-shot	0.528	0.510	1.000	+0.21
		LX-bg + TEA	0.528	0.532	1.000	-0.18
		RE + TEA	0.528	0.503	1.000	+0.22
		RPX (random) + TEA	0.528	0.190	<0.001	+0.92

Table 6: Deadline-conditioned localization on HAI 21.03 (T = with TEA). Latency: median measured per explanation during the runs (1 thread, pooled over detectors); Evals: detector evaluations per explanation.

Method	DCL@1(D)				DCL@3	Lat. [ms]	Evals
	30 s T	60 s	60 s T	120 s T	60 s T		
RE	0.46	0.46	0.50	0.51	0.68	0.5	1
SM	0.32	0.31	0.32	0.39	0.49	1.0	1
IG	0.51	0.49	**0.54**	**0.63**	0.68	3.4	32
KernelSHAP-eq	0.47	0.41	0.48	0.55	0.67	20.4	512
KernelSHAP-1024	0.49	0.46	0.51	0.59	0.68	40.9	1024
LIME-eq	0.50	0.48	0.52	0.60	0.68	38.1	512
LIME-1024	0.51	0.50	0.52	0.60	0.69	85.9	1024
Fung ens.	**0.54**	**0.51**	**0.54**	0.62	**0.72**	43.8	514
CLX (clusters)	0.30	0.34	0.32	0.34	0.61	14.2	512
RPX (random)	0.16	0.17	0.19	0.17	0.52	13.0	512
LX-bg	0.51	**0.51**	0.53	0.60	0.70	200.2	4096
LX (ours)	0.51	**0.51**	0.53	0.60	0.70	28.2	512

Table 7: Deadline-conditioned localization on PCL-Sim (T = with TEA). Latency: median measured per explanation during the runs (1 thread, pooled over detectors); Evals: detector evaluations per explanation. In-time: hazard deadline minus a 30 s operator reserve, hazardous attacks only.

Method	DCL@1(D)				DCL@3	In-time@1		In-time@3	Lat. [ms]	Evals
	30 s T	60 s	60 s T	120 s T	60 s T	single	T	T		
RE	0.49	0.21	0.49	0.50	**0.53**	0.17	0.41	**0.54**	0.3	1
SM	0.47	0.21	0.46	0.47	**0.53**	0.19	0.40	**0.54**	0.8	1
IG	**0.51**	**0.34**	**0.52**	**0.52**	**0.53**	**0.33**	**0.50**	**0.54**	2.2	32
KernelSHAP-eq	0.40	0.23	0.40	0.40	0.48	0.25	0.40	0.51	2.4	64
KernelSHAP-1024	0.48	0.24	0.48	0.48	0.52	0.22	0.43	0.51	24.2	1024
LIME-eq	0.42	0.21	0.42	0.42	0.50	0.20	0.38	0.48	1.8	64
LIME-1024	0.50	0.24	0.50	0.50	**0.53**	0.22	0.46	0.52	15.4	1024
Fung ens.	0.50	0.23	0.50	0.50	**0.53**	0.21	0.44	**0.54**	3.2	66
CLX (clusters)	0.10	0.09	0.10	0.10	0.41	0.10	0.14	0.38	1.6	64
RPX (random)	0.32	0.17	0.33	0.35	0.51	0.17	0.29	0.50	1.5	64
LX-bg	**0.51**	0.28	**0.52**	**0.52**	**0.53**	0.29	0.49	**0.54**	7.3	512
LX (ours)	**0.51**	0.27	**0.52**	0.51	**0.53**	0.28	0.48	0.53	1.6	64

Table 8: Explanation latency on one CPU thread (Intel Xeon at 2.1 GHz, 4-core cloud VM; short unrelated jobs ran concurrently during part of the PCL-Sim run) (median over 10 incidents × 5 repetitions, ms; ShaTS: 2 incidents, 1 repetition). Evals: detector evaluations per explanation.

Data	Method	Evals	Linear	MLP	CNN	GRU
PCL-Sim	RE	1	0.1	0.1	0.3	0.6
	SM	1	0.4	0.6	0.9	3.7
	IG	32	0.5	1.0	3.1	5.7
	KernelSHAP-eq	64	1.2	1.5	3.1	2.8
	KernelSHAP-1024	1024	9.4	12.5	35.1	28.7
	LIME-eq	64	0.4	0.8	2.6	2.4
	LIME-1024	1024	2.0	5.1	26.6	21.0
	Fung ens.	514	1.9	3.7	15.4	15.1
	LX-bg	512	0.9	2.6	13.6	9.5
	ShaTS-sensor	19360	109.9	177.9	612.5	471.6
	LX (ours)	64	0.3	0.7	2.5	2.1
	H-LX (tags)	16	0.6	1.1	3.5	3.3
HAI 21.03	RE	1	0.1	0.2	0.3	0.7
	SM	1	0.4	0.7	0.9	3.7
	IG	32	0.7	1.4	3.5	6.1
	KernelSHAP-eq	512	6.8	10.2	20.9	18.7
	KernelSHAP-1024	1024	12.7	19.4	42.7	38.6
	LIME-eq	512	2.7	6.3	17.5	14.8
	LIME-1024	1024	4.6	12.0	33.5	31.2
	Fung ens.	514	3.3	7.6	18.6	19.7
	LX-bg	4096	14.1	44.3	165.0	185.2
	ShaTS-sensor	129960	1021.6	1871.4	4790.0	4384.8
	LX (ours)	512	2.2	5.9	16.9	13.7
	H-LX (tags)	64	2.8	6.6	20.0	17.2

Table 9: Explanation quality. Temporal agreement: share of consecutive explanations (5 s apart, first 120 s) with the same top-1 loop. Noise Jaccard: top-3 overlap under Gaussian input noise ($\sigma = 0.1$) at alarm and +60 s. Deletion: area under the normalised score curve when loops are restored to the training mean in ranked order (lower is better), at the alarm. Tag HR@k: rank of a manipulated tag among all tags at the alarm (for LX: H-LX refinement); HAI excludes P2 setpoint targets that are invisible in the data.

Data	Method	Temporal top-1 agr.		Noise Jacc.@3	Deletion↓	Tag HR@1	Tag HR@3
		single	TEA				
PCL-Sim	RE	0.53	0.99	0.49	0.18	0.38	**0.76**
	SM	0.52	0.99	0.53	0.28	0.20	0.53
	IG	0.71	**1.00**	0.53	0.18	0.44	0.74
	KernelSHAP-eq	**0.77**	0.99	0.71	0.19	0.43	0.69
	KernelSHAP-1024	0.74	0.99	0.62	0.18	0.44	0.73
	LIME-eq	0.74	0.99	0.71	0.21	0.41	0.69
	LIME-1024	0.70	0.99	0.59	0.20	0.44	0.73
	Fung ens.	0.59	0.99	0.60	**0.17**	0.28	0.75
	CLX (clusters)	0.61	0.99	0.64	0.34	0.05	0.27
	RPX (random)	0.63	0.98	**0.76**	0.26	0.00	0.08
	LX-bg	0.66	0.99	0.55	0.22	–	–
	LX (ours) (H-LX)	0.64	0.99	0.51	0.21	**0.45**	**0.76**
HAI 21.03	RE	0.58	0.95	0.59	0.18	0.23	0.40
	SM	0.62	0.95	0.44	0.45	0.10	0.34
	IG	0.67	**0.97**	0.59	0.17	0.28	0.41
	KernelSHAP-eq	**0.69**	**0.97**	0.73	0.19	0.29	0.39
	KernelSHAP-1024	0.68	**0.97**	0.70	0.18	0.31	0.41
	LIME-eq	0.67	**0.97**	0.65	0.17	0.30	0.41
	LIME-1024	0.68	**0.97**	0.62	0.17	0.31	0.41
	Fung ens.	0.66	0.96	0.54	**0.16**	**0.36**	0.48
	CLX (clusters)	**0.69**	0.96	0.75	0.26	0.00	0.00
	RPX (random)	0.63	0.95	**0.76**	0.33	0.00	0.00
	LX-bg	0.68	0.96	0.57	**0.16**	–	–
	LX (ours) (H-LX)	0.68	0.96	0.56	0.17	0.31	**0.49**

Table 10: Ablations and robustness (HR@1 unless noted; "alarm": single shot at the alarm; "+60 T": 60 s after the alarm with TEA). Reference-gap and noise rows: CNN detector, seed 0. Noise rows: Gaussian noise of standard deviation σ_n (standardised units) added to the test and validation data, with σ and θ recalibrated on the noisy validation data (model weights frozen). Decoy rows: linear and CNN detectors, 3 seeds; decoy@1 is the share of attacks in which the innocent decoy loop is ranked first.

Data	Condition	LX		IG		KS-1024		RE	
		alarm	+60 T	alarm	+60 T	alarm	+60 T	alarm	+60 T
HAI	Δ_{ref}=32 s	0.67	0.67	0.64	0.67	0.64	0.71	0.73	0.69
HAI	Δ_{ref}=60 s (default)	0.73	0.67	0.64	0.67	0.69	0.69	0.73	0.69
HAI	Δ_{ref}=120 s	0.71	0.64	0.69	0.67	0.69	0.67	0.73	0.69
HAI	Δ_{ref}=300 s	0.64	0.62	0.64	0.71	0.64	0.71	0.73	0.69
HAI	σ_n=0.1, recal.	0.64	0.61	0.61	0.70	0.66	0.68	0.64	0.73
HAI	σ_n=0.3, recal.	0.51	0.56	0.56	0.60	0.56	0.63	0.56	0.44
PCL-Sim	Δ_{ref}=32 s	0.79	0.76	0.78	0.75	0.78	0.78	0.82	0.79
PCL-Sim	Δ_{ref}=60 s (default)	0.80	0.76	0.80	0.77	0.79	0.78	0.82	0.79
PCL-Sim	Δ_{ref}=120 s	0.80	0.76	0.82	0.78	0.82	0.79	0.82	0.79
PCL-Sim	Δ_{ref}=300 s	0.81	0.78	0.82	0.78	0.82	0.78	0.82	0.79
PCL-Sim	σ_n=0.1, recal.	0.93	0.91	0.93	0.92	0.93	0.93	0.94	0.91
PCL-Sim	σ_n=0.3, recal.	0.94	0.93	0.95	0.95	0.95	0.94	0.95	0.93
PCL-Sim	no decoy: HR@1	0.84	0.80	0.84	0.82	0.83	0.77	0.85	0.78
PCL-Sim	decoy: HR@1	0.86	0.81	0.85	0.83	0.84	0.78	0.85	0.79
PCL-Sim	decoy: decoy@1	0.03	0.05	0.03	0.04	0.03	0.05	0.04	0.05

Table 11: Outcome of the pre-specified hypotheses.

H	Statement	Outcome
H1	LX localizes better than tag-level methods	partially: best black-box at the alarm on HAI; not better than IG, nor than LX-bg on PCL-Sim
H2	TEA improves accuracy after the alarm and temporal stability	supported (both benchmarks, all methods)
H3	LX+TEA has deterministic cost and the best DCL / in-time rate	cost: supported; DCL: not supported (tied on HAI; IG/LX-bg slightly better on PCL-Sim)
H4	control structure beats data-driven / random grouping	supported ($p < 10^{-10}$)
H5a	LX+TEA degrades less under sensor noise	not supported (no significant differences; LX numerically lower on HAI)
H5b	decoy captures top-1 less often with LX+TEA	not supported (decoy ineffective against all methods, $\leq$5 % with TEA)
H6	random splits inflate detection	supported (F1 0.940 vs. 0.575)

Table B.12: Validation protocols stated in reviewed XAI-IDS studies that use time-series CPS/IIoT data (as stated in the papers; "–" if not stated in the sections we read). Random proportional splits or k-fold CV without a stated chronological or gapped protocol allow overlapping or adjacent windows on both sides of the split.

Study	Data	Stated protocol
Ha et al. (2022)	gas pipeline	80/20 split
Yazdinejad et al. (2025)	SWaT, gas pipeline, UNSW-NB15	70/30 split
Birihanu and Lendák (2024)	SWaT, HAI, Modbus	80/20 split
Xu et al. (2024)	SWaT, WADI	five-fold CV
Rasheed et al. (2026)	SWaT, WADI, BATADAL, IIoT	5-fold CV, 70/15/15
Panesar et al. (2026)	HIL microgrid scalograms	70/30 split
Shoukat et al. (2025)	N-BaIoT, Edge-IIoTset	70/30 split
Attique et al. (2024)	CICIDS2017, X-IIoTID	70/30 split
Saiyed et al. (2026)	WUSTL-IIoT-2021, CICIoT2023	stratified 5-fold CV
Franco de la Peña et al. (2026)	SWaT	padded segment split (leakage-aware)
Fung et al. (2024)	SWaT, WADI, TEP	normal-only training, attack test sets

Table D.13: Loop HR@1 per detector family ("at alarm, single-shot / 60 s after the alarm with TEA"; mean over attacks of the per-attack mean over 3 seeds; n: attacks detected by at least one seed).

Data	Detector	n	RE	IG	KernelSHAP-1024	Fung ens.	LX-bg	LX
HAI	linear	46	0.70 / 0.66	0.85 / 0.78	0.82 / 0.72	0.87 / 0.82	0.85 / 0.80	0.85 / 0.80
	mlp	45	0.75 / 0.64	0.71 / 0.70	0.70 / 0.70	0.73 / 0.72	0.71 / 0.68	0.75 / 0.68
	cnn	45	0.73 / 0.60	0.67 / 0.69	0.70 / 0.68	0.70 / 0.70	0.70 / 0.65	0.75 / 0.67
	gru	45	0.66 / 0.53	0.76 / 0.73	0.74 / 0.65	0.72 / 0.64	0.70 / 0.70	0.81 / 0.63
PCL-Sim	linear	270	0.89 / 0.80	0.89 / 0.87	0.87 / 0.76	0.88 / 0.82	0.89 / 0.86	0.89 / 0.84
	mlp	226	0.87 / 0.85	0.87 / 0.86	0.86 / 0.85	0.90 / 0.87	0.85 / 0.86	0.85 / 0.83
	cnn	253	0.80 / 0.76	0.79 / 0.76	0.78 / 0.78	0.81 / 0.76	0.83 / 0.79	0.79 / 0.76
	gru	265	0.87 / 0.81	0.88 / 0.87	0.86 / 0.81	0.88 / 0.84	0.89 / 0.88	0.89 / 0.86

Table D.14: Tag-level localization (HR@1 / HR@3, single-shot at the given time after the alarm). H-LX refines the top loop of LX into tags; the other methods rank all tags directly.

Data	Method	+0 s	+30 s	+60 s	+120 s
HAI	RE	0.23 / 0.40	0.13 / 0.27	0.13 / 0.27	0.02 / 0.11
	SM	0.10 / 0.34	0.13 / 0.32	0.13 / 0.26	0.10 / 0.20
	IG	0.28 / 0.41	0.13 / 0.27	0.15 / 0.29	0.07 / 0.15
	KS-1024	0.31 / 0.41	0.15 / 0.29	0.16 / 0.27	0.08 / 0.17
	LIME-1024	0.31 / 0.41	0.15 / 0.28	0.17 / 0.29	0.06 / 0.16
	Fung-ens	0.36 / 0.48	0.23 / 0.37	0.22 / 0.36	0.10 / 0.19
	H-LX	0.31 / 0.49	0.12 / 0.34	0.15 / 0.31	0.06 / 0.18
PCL-Sim	RE	0.38 / 0.76	0.23 / 0.37	0.26 / 0.43	0.26 / 0.42
	SM	0.20 / 0.53	0.18 / 0.37	0.17 / 0.38	0.16 / 0.37
	IG	0.44 / 0.74	0.21 / 0.42	0.25 / 0.47	0.22 / 0.42
	KS-1024	0.44 / 0.73	0.14 / 0.31	0.19 / 0.37	0.18 / 0.33
	LIME-1024	0.44 / 0.73	0.15 / 0.32	0.21 / 0.40	0.20 / 0.36
	Fung-ens	0.28 / 0.75	0.14 / 0.36	0.19 / 0.41	0.18 / 0.38
	H-LX	0.45 / 0.76	0.17 / 0.32	0.21 / 0.39	0.19 / 0.36

Table D.15: PCL-Sim loop HR@1 by attack primitive (at alarm, single-shot / +60 s with TEA; n: detected (attack, detector) units).

Primitive	n	RE	IG	LX-bg	LX
CV	135	0.87 / 0.82	0.86 / 0.84	0.87 / 0.82	0.88 / 0.83
CV_PV	160	0.96 / 0.91	0.95 / 0.88	0.95 / 0.88	0.95 / 0.87
CV_ST	159	0.94 / 0.96	0.93 / 0.96	0.94 / 0.98	0.93 / 0.96
PV_BIAS	148	0.93 / 0.85	0.94 / 0.95	0.93 / 0.94	0.93 / 0.93
PV_FREEZE	115	0.56 / 0.54	0.51 / 0.49	0.56 / 0.54	0.50 / 0.46
SP	137	0.86 / 0.82	0.90 / 0.89	0.90 / 0.87	0.90 / 0.86
SP_PV	160	0.79 / 0.66	0.81 / 0.80	0.86 / 0.81	0.81 / 0.76

Table D.16: HAI 21.03 loop HR@1 by manipulated variable type, read from the target-controller code of the official attack table (single-shot at the alarm / +60 s with TEA; pooled over 4 detectors). Strata are small; descriptive only.

Manipulated variable	attacks	RE	IG	KernelSHAP-1024	LX-bg	LX
SP only	8	0.69 / 0.38	0.66 / 0.52	0.66 / 0.46	0.65 / 0.47	0.73 / 0.46
SP + PV spoof	4	0.50 / 0.31	0.56 / 0.65	0.58 / 0.60	0.65 / 0.62	0.71 / 0.62
CV only	8	0.69 / 0.74	0.79 / 0.79	0.79 / 0.78	0.79 / 0.84	0.79 / 0.79
CV + PV spoof	3	0.94 / 0.89	0.97 / 0.92	0.92 / 0.78	0.97 / 0.94	1.00 / 0.89
combined (2 scenarios)	23	0.73 / 0.66	0.77 / 0.76	0.75 / 0.74	0.74 / 0.73	0.80 / 0.74